

DIALECTA

Don't think alone. Carry a team.

A real-time, multimodal, adversarial multi-agent system that acts as a **cognitive parliament** in your pocket. Four AI agents, each with a distinct thinking style, debate your decisions as you make them — catching cognitive biases before they cost you.

The Problem

- Cognitive bias is estimated to cost the global economy **\$1.5 trillion annually** through bad financial decisions, poor hires, and failed projects.
- Everyone has made a decision they regretted within 24 hours — and almost no one has a system that catches it *before* it happens.
- Most AI assistants are reactive: you ask, it answers. DIALECTA is proactive — it intercepts you at the point of commitment.

Core Architecture: The Cognitive Parliament

Four agents, four jobs, one live debate:

Agent	Role
The Optimist	Surfaces opportunities, finds supporting evidence, builds the best-case scenario
The Skeptic	Stress-tests assumptions, hunts for flaws, demands evidence, asks “what could go wrong?”
The Analyst	Crunches the numbers, checks historical patterns, runs quick projections
The Ethicist	Tests alignment against stated values, long-term goals, and broader impact

The agents don't just answer — they argue with each other, live.

Tech Stack

Layer	Technology	Purpose
Multi-Agent Orchestration	CrewAI / LangGraph	Custom task delegation across the four personas
Context Integration	Custom MCP servers	Browser, email, calendar, documents — real user data
Multimodal Input	Whisper + Vision	Voice and screenshot/webcam context, not text-only
Real-Time Streaming	WebRTC + WebSocket	Live, low-latency voice debates

Layer	Technology	Purpose
Edge / Local Processing	Llama 3.1/3.2 or Qwen (local)	Privacy-preserving inference where possible
Intervention Layer	Browser extension + mobile + wearable	Meets the user at the actual decision point

How It Works (Demo Flow)

- **The Hook** — a familiar moment: about to commit to a decision you might regret.
- **The Intercept** — at checkout, the Analyst flags unused past purchases, the Skeptic calls out fake urgency in a “limited time” banner, and the Ethicist checks it against a stated savings goal.
- **The Voice Debate** — ask out loud, e.g. “Should I send this angry email to my client?” — and the four agents debate it live, citing your real sent folder and calendar.
- **The Decision** — you decide, fully informed, in seconds.

Why This Wins

- **Universal, high-pain problem** — every judge has personally made a bad decision in the heat of the moment.
- **Rides the right technical waves** — agentic AI, MCP-based tool integration, multimodal input, and adversarial multi-agent design.
- **Technical depth, not a wrapper** — four coordinated agent personalities with live tool access and real-time voice.
- **Clear differentiation** — not a chatbot, not a coding assistant, not RAG search: a new category, personal cognitive infrastructure.
- **A real business underneath** — freemium for everyday use, premium for high-stakes decisions, B2B compliance angle for regulated industries.

Implementation Roadmap (48-Hour Build)

Phase 1 — MVP Core (Hours 0–12)

- [] Stand up the four agent personas in CrewAI/LangGraph
- [] Build one MCP server (browser extension for web decisions)
- [] Create the debate output formatter

Phase 2 — Polish & Demo (Hours 12–30)

- [] Add voice input via streaming Whisper
- [] Add vision input via screenshot context
- [] Build the browser extension UI around the intervention moment
- [] Pre-seed 3–5 compelling demo scenarios

Phase 3 — Presentation (Hours 30–48)

- [] Record a 3-minute demo video (purchase intervention + email intervention)

- [] Build the slide deck around the cognitive parliament metaphor
- [] Clean up the GitHub repo: README, architecture diagram, setup instructions

Theme Variations

Theme	Pivot
Healthcare	Clinical Dialecta — debates treatment decisions against patient history
Finance	Trade Parliament — real-time trade analysis with risk, compliance, opportunity agents
Education	Socratic Council — students debate against Socratic / devil's-advocate AI personas
Sustainability	Green Council — evaluates purchases against carbon footprint and ethical sourcing
Accessibility	NeuroParliament — tuned for ADHD/autism/executive dysfunction, task follow-through

Business Model

- **Freemium** — everyday decisions are free; deep financial/medical decision support is a paid tier.
- **B2B** — enterprise decision-making compliance, a natural fit for banks, insurers, and other regulated industries needing an auditable decision trail.

Pitch Script

“When you shop online, an algorithm built by someone else is optimizing against you. When you make decisions, you’re alone with your biases. DIALECTA is a real-time cognitive immune system — four AI advisors with access to your real data, debating every important choice before you commit. It’s not a chatbot. It’s not a search engine. It’s a cognitive parliament in your pocket. And it was built in 48 hours.”

Status: in active development — built for hackathon submission.
DIALECTA — Project Brief